

LAB 01

Aim: To perform and understand fundamental DBMS operations, including database creation, table management, data manipulation, modification, and retrieval using MySQL.

SHOW databases;

```
No default schema selected; type \use <schema> to set one.
MySQL localhost:3307 ssl SQL > show databases
-> ;
```

Database
information_schema
mysql
performance_schema
pranavdubey16_test
sys

```
5 rows in set (0.0012 sec)
```

CREATE DATABASE stud1_test;

SHOW DATABASES;

```
MySQL localhost:3307 ssl SQL > create database parthSonarD6;
Query OK, 1 row affected (0.0780 sec)
MySQL localhost:3307 ssl SQL > show databases
-> ;
```

Database
information_schema
mysql
parth_sonar_d6
parthsonard6
performance_schema
pranavdubey16_test
sys

```
7 rows in set (0.0011 sec)
```

CREATE DATABASE stud1test;

SHOW DATABASES;

```
MySQL localhost:3307 ssl SQL> create database Parth_Sonar_D6
-> ;
Query OK, 1 row affected (0.0791 sec)
MySQL localhost:3307 ssl SQL> show databases
-> ;
```

Database
information_schema
mysql
parth_sonar_d6
performance_schema
pranavdubey16_test
sys

DROP DATABASE stud1test;

SHOW databases;

```
MySQL localhost:3307 ssl SQL> drop database parthsonard6
-> ;
Query OK, 0 rows affected (0.1051 sec)
MySQL localhost:3307 ssl SQL> show databses
-> ;
ERROR: 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'databses' at line 1
MySQL localhost:3307 ssl SQL> show databases;
```

Database
information_schema
mysql
parth_sonar_d6
performance_schema

USE stud1_test;

CREATE TABLE Persons (

PersonID int,

LastName varchar(25),

FirstName varchar(25),

Address varchar(100),

City varchar(25)

);

SHOW TABLES;

DESCRIBE Persons;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > use parth_sonar_d6;
Default schema set to `parth_sonar_d6`.
Fetching global names, object names from `parth_sonar_d6` for auto-completion... Press ^C to stop.
MySQL localhost:3307 ssl parth_sonar_d6 SQL > create table persons(personID int,LastName varchar(25),FirstName varchar(25),Address varchar(100),city varchar(25));
Query OK, 0 rows affected (0.2969 sec)
MySQL localhost:3307 ssl parth_sonar_d6 SQL > show tables;
+-----+
| Tables_in_parth_sonar_d6 |
+-----+
| persons                   |
+-----+
1 row in set (0.0038 sec)
```

DROP TABLE Persons;

```
ERROR 1051 (42502): UNKNOWN TABLE 'parth_sonar_d6.persons'
MySQL localhost:3307 ssl parth_sonar_d6 SQL > drop table persons;
Query OK, 0 rows affected (0.1810 sec)
```

CREATE TABLE Customers (CustomerID int,LastName varchar(25),FirstName

varchar(25),Address varchar(100),City varchar(25), primary key

(CustomerID));

DESCRIBE Customers;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > create table Customers(customerID int,LastName varchar(25),FirstName varchar(25),Address varchar(100),city varchar(25),primary key(CustomerID));
Query OK, 0 rows affected (0.3265 sec)
MySQL localhost:3307 ssl parth_sonar_d6 SQL > show tables;
+-----+
| Tables_in_parth_sonar_d6 |
+-----+
| customers                 |
+-----+
1 row in set (0.0013 sec)
```

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > describe customers
-> ;
```

Field	Type	Null	Key	Default	Extra
customerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
city	varchar(25)	YES		NULL	

5 rows in set (0.0025 sec)

INSERT INTO Customers

(CustomerID,LastName,FirstName,Address,City) VALUES

(96365,'Shah','Hiren','A67,Building10,abc lane','Mumbai');

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > insert into customers(customerID,LastName,FirstName,Address,city) values(1,'sonar','parth','street vivekanand colony',
'mumbai');
Query OK, 1 row affected (0.0834 sec)
```

SELECT * FROM Customers;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > select * from customers;
+-----+-----+-----+-----+-----+
| customerID | LastName | FirstName | Address | city |
+-----+-----+-----+-----+-----+
| 1 | sonar | parth | street vivekanand colony | mumbai |
+-----+-----+-----+-----+-----+
1 row in set (0.0005 sec)
```

INSERT INTO Customers VALUES

(96300,'Singh','Prakesh','B7,House10,abc lane','Mumbai'),

(89790,'Joshi','Ramesh','413,Sadashiv Nagar','Pune');

----Now add approx. 10 customers----

--SELECT * FROM Customers;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > select*from customers;
```

customerID	LastName	FirstName	Address	city
1	sonar	parth	street vivekanand colony	mumbai
2	lane	adam	new york street st7007	new york
3	shane	tony	new jersy street	new jersy
4	lane	adam	new york street st7007	new york
5	shane	tony	new jersy street	new jersy
6	goone	lane	amsterdam street	georgiana
7	casey	cowboy	china town	london
8	doom	victor	latveria	earth

8 rows in set (0.0006 sec)

ALTER TABLE Customers ADD Email varchar(50);

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > alter table customers add emailid varchar(25);
Query OK, 0 rows affected (0.2478 sec)
```

Records: 0 Duplicates: 0 Warnings: 0

DESCRIBE Customers;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > describe customers
-> ;
```

Field	Type	Null	Key	Default	Extra
customerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
city	varchar(25)	YES		NULL	
emailid	varchar(25)	YES		NULL	

6 rows in set (0.0015 sec)

ALTER TABLE Customers ADD Remark varchar(255);

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > alter table customers add remark varchar(25);
Query OK, 0 rows affected (0.2138 sec)
```

Records: 0 Duplicates: 0 Warnings: 0

DESCRIBE Customers;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > describe customers;
```

Field	Type	Null	Key	Default	Extra
customerID	int	NO	PRI	NULL	
LastName	varchar(25)	YES		NULL	
FirstName	varchar(25)	YES		NULL	
Address	varchar(100)	YES		NULL	
city	varchar(25)	YES		NULL	
emailid	varchar(25)	YES		NULL	
remark	varchar(25)	YES		NULL	

7 rows in set (0.0016 sec)

SELECT * FROM Customers;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > select*from customers;
```

customerID	LastName	FirstName	Address	city	emailid	remark
1	sonar	parth	street vivekanand colony	mumbai	NULL	NULL
2	lane	adam	new york street st7007	new york	NULL	NULL
3	shane	tony	new jersey street	new jersey	NULL	NULL
4	lane	adam	new york street st7007	new york	NULL	NULL
5	shane	tony	new jersey street	new jersey	NULL	NULL
6	goone	lane	amsterdam street	georgiana	NULL	NULL
7	casey	cowboy	china town	london	NULL	NULL
8	doom	victor	latveria	earth	NULL	NULL

8 rows in set (0.0005 sec)

UPDATE Customers SET Email='ramesh.joshi@abc.com' , Remark='Regular Buyer' WHERE

CustomerID=89790;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > update customers set emailid='sonarparth@gamil.com',remark='valid' where customerid=1;
Query OK, 1 row affected (0.0605 sec)

Rows matched: 1  Changed: 1  Warnings: 0
```

SELECT CustomerID,Address FROM Customers;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > select customerid,address from customers;
```

customerid	address
1	street vivekanand colony
2	new york street st7007
3	new jersy street
4	new york street st7007
5	new jersy street
6	amsterdam street
7	china town
8	latveria

```
8 rows in set (0.0005 sec)
```

----Try yourself on different set of attributes-----

SELECT CustomerID,Address FROM Customers WHERE CustomerID=96300;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > select customerid,address,firstname from customers where customerid=1;
```

customerid	address	firstname
1	street vivekanand colony	parth

```
1 row in set (0.0007 sec)
```

SELECT * FROM Customers WHERE CustomerID=96300;

```
MySQL localhost:3307 ssl parth_sonar_d6 SQL > select * from customers where customerid=8;
```

customerID	LastName	FirstName	Address	city	emailid	remark
8	doom	victor	latveria	earth	NULL	NULL

```
1 row in set (0.0005 sec)
```

What is a database? Why do we need a database management system such as MySQL?

Database is an organized collection of Data which stored electronically for easy access, updating, and deleting data. We need Database Management System as it is secure, scalable, organisable and can retrieve data easy that's why we need the Database Management Systems.

What is the difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE?

- i) Create database: This Command is to create new database in mysql Database management System
- ii) Show Database: This Command is to show the all databases there are in Database Management System the server Host
- iii) Drop Database: This command is used to delete already existing database.

What is a table in a database? Explain the purpose of rows and columns in a table.

In a relational database, a table is a structured collection of data organized into a grid of horizontal rows and vertical columns. It serves as the core data storage unit where all information regarding a specific subject (such as users, orders, or products) is neatly kept

In a relational database, rows and columns are the fundamental building blocks of a table used to organize and store data. A row represents a single, complete record (such as an individual customer), while a column defines a specific attribute shared across all records (such as an email address).

What is a primary key? Why was CustomerID selected as the primary key in the Customers table?

Primary Key refers to the the the entity in relational database which is uniquely identifies every row in the table of database.

Customer ID is the primary key in Customer table because it uniquely identities each rows of this table.

What is the purpose of the VARCHAR and INT data types? Give one suitable example of each.

In MYSQL , varchar and int are the distinct data types Used to define what kind of data table coloumn can store.

For e.g.:

Create Table persons(Address varchar(100));

Create Table persons(personid int);

What is the difference between INSERT, SELECT, UPDATE, and ALTER TABLE?

i) Insert : Creates brand new entries. Use this when adding a new user, a new product, or a new log entry to your database.

Example: INSERT INTO users (name, age) VALUES ('Alice', 25);

ii) Select:Fetches and views records without altering them. You can filter results, sort them, or join multiple tables together.

Example: SELECT * FROM users WHERE age > 20;

Why is the WHERE clause important in an UPDATE statement? What could happen if we write an UPDATE statement without a WHERE condition?

The WHERE clause is important in an UPDATE statement because it specifies which rows should be changed.

For example:

UPDATE students

SET marks = 90

WHERE student_id = 101;

This updates the marks only for student 101.

If you write an UPDATE statement without a WHERE clause, the database will update every row in the table.

For example:

```
UPDATE students  
SET marks = 90;
```

This would set the marks to 90 for all students, which could cause unintended data loss or incorrect data.